

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12419114
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Li; Jian-Sing et al.

Integrated circuit

Abstract

An integrated circuit includes a first transistor of a first conductivity type including a first active area extending in a first direction; a second transistor of the first conductivity type including at least two second active areas extending in the first direction and a first gate stripe crossing the at least two second active areas; and a third transistor of a second conductivity type that is stacked on the second transistor and includes at least two third active areas arranged above the at least two second active areas. A top most boundary line of the first active area is aligned with a top most boundary line of one of the at least two third active areas in a layout view.

Inventors: Li; Jian-Sing (Hsinchu, TW), Wu; Guo-Huei (Tainan, TW), Zhuang; Hui-Zhong (Kaohsiung, TW), Chen; Chih-Liang (Hsinchu, TW), Tien; Li-Chun (Tainan, TW)

Applicant: TAIWAN SEMICONDUCTOR MANUFACTURING CO., LTD. (Hsinchu, TW)

Family ID: 1000008820205

Assignee: TAIWAN SEMICONDUCTOR MANUFACTURING CO., LTD. (Hsinchu, TW)

Appl. No.: 18/472985

Filed: September 22, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240014203 A1	Jan. 11, 2024

Related U.S. Application Data

continuation parent-doc US 17406663 20210819 US 11830869 child-doc US 18472985
continuation parent-doc US 16837970 20200401 US 11107805 child-doc US 17406663
us-provisional-application US 62834141 20190415

Publication Classification

Int. Cl.: **H01L27/02** (20060101); **G06F30/392** (20200101); **H01L21/02** (20060101); **H01L21/8238** (20060101); **H01L27/092** (20060101); **H01L29/06** (20060101); **H01L29/423** (20060101); **H01L29/786** (20060101); **H10D30/67** (20250101); **H10D62/10** (20250101); **H10D84/01** (20250101); **H10D84/03** (20250101); **H10D84/85** (20250101); **H10D89/10** (20250101)

U.S. Cl.:

CPC **H10D89/10** (20250101); **G06F30/392** (20200101); **H01L21/02603** (20130101); **H10D30/6735** (20250101); **H10D30/6757** (20250101); **H10D62/121** (20250101); **H10D84/0167** (20250101); **H10D84/0186** (20250101); **H10D84/0193** (20250101); **H10D84/038** (20250101); **H10D84/856** (20250101);

Field of Classification Search

CPC: H01L (27/0207); H01L (21/02603); H01L (21/823807); H01L (21/823821); H01L (21/823871); H01L (27/0922); H01L (29/0673); H01L (29/42392); H01L (29/78696); H01L (27/11807); H01L (21/8221); H01L (27/0924); H01L (2027/11874); H01L (27/0688); H01L (27/092); H01L (27/088); H01L (21/8238); H01L (21/823828); H01L (29/66772); H01L (29/78654); G06F (30/392)

USPC: 257/351

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10157921	12/2017	Chung	N/A	N/A
11107805	12/2020	Li	N/A	N/A
11830869	12/2022	Li	N/A	H01L 29/42392
2014/0245248	12/2013	Yang	N/A	N/A
2015/0061026	12/2014	Lin	257/369	H10D 89/10
2017/0194349	12/2016	Chen et al.	N/A	N/A
2018/0158811	12/2017	Subhash et al.	N/A	N/A
2019/0006392	12/2018	Liu et al.	N/A	N/A
2020/0104446	12/2019	Li	N/A	N/A
2020/0235013	12/2019	Lilak	N/A	N/A
2020/0328210	12/2019	Wu	N/A	H01L 23/5384

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
20040016403	12/2003	KR	N/A
20170124940	12/2016	KR	N/A
I566365	12/2016	TW	N/A
201834185	12/2017	TW	N/A
201903859	12/2018	TW	N/A

OTHER PUBLICATIONS

More Moore Team, "International roadmap for devices and systems 2017 edition", IEEE, 2018. cited by applicant

J. Ryckaert et al., "The Complementary FET (CFET) for CMOS scaling beyond N3", 2018 IEEE Symposium on VLSI Technology, 2018. cited by applicant

Sorin Dobre et al., "Mixed Cell-Height Implementation for Improved Design Quality in Advanced Nodes", 2015 IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2015. cited by applicant

Jianli Chen et al., "Mixed-Cell-Height Placement with Complex Minimum-Implant-Area Constraints", 2018 IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2018. cited by applicant

Primary Examiner: Kim; Tong-Ho

Attorney, Agent or Firm: Maschoff Brennan

Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a continuation of U.S. application Ser. No. 17/406,663, filed Aug. 19, 2021, which is a continuation of U.S. application Ser. No. 16/837,970, filed Apr. 1, 2020, now U.S. Pat. No. 11,107,805, issued Aug. 31, 2021, which claims priority of U.S. Provisional Application Ser. No. 62/834,141, filed Apr. 15, 2019, which is herein incorporated by reference.

BACKGROUND

(1) Along with the progressive process of manufacturing semiconductor devices, layout design with cells of different cell heights has been widely implemented in the industry. Furthermore, greater cell density with mixed cell height in the layout and better computing performance of cells within limited area is demanded.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is an equivalent circuit of part of an integrated circuit, in accordance with various embodiments.

(3) FIG. 2A is a perspective diagram of part of the integrated circuit of FIG. 1, in accordance with various embodiments. FIG. 2B is a layout diagram in a plan view of part of the integrated circuit corresponding to a part of FIG. 2A, in accordance with various embodiments.

(4) FIG. 3A is a perspective diagram of part of an integrated circuit, in accordance with various embodiments. FIG. 3B is a layout diagram in a plan view of part of the integrated circuit corresponding to a part of FIG. 3A, in accordance with various embodiments. FIG. 3C is a layout diagram in a plan view of part of the integrated circuit corresponding to a part of FIG. 3A, in accordance with various embodiments. FIG. 3D is a layout diagram in a plan view of part of the integrated circuit corresponding to a part of FIG. 3A, in accordance with various embodiments.

(5) FIG. 4 is a layout diagram in a plan view of part of an integrated circuit, in accordance with

various embodiments.

(6) FIG. 5 is a layout diagram in a plan view of part of an integrated circuit, in accordance with various embodiments.

(7) FIG. 6 is a layout diagram in a plan view of part of an integrated circuit, in accordance with various embodiments.

(8) FIG. 7 is a layout diagram in a plan view of part of an integrated circuit, in accordance with various embodiments.

(9) FIG. 8 is a layout diagram in a plan view of part of an integrated circuit, in accordance with various embodiments.

(10) FIG. 9 is a layout diagram in a plan view of part of an integrated circuit, in accordance with various embodiments.

(11) FIG. 10 is a layout diagram in a plan view of part of an integrated circuit, in accordance with various embodiments.

(12) FIG. 11 is a flow chart of a method of generating a layout design for fabricating an integrated circuit, in accordance with some embodiments of the present disclosure.

(13) FIG. 12 is a flow chart of a method of fabricating an integrated circuit based on the layout design generated by the method of FIG. 11, in accordance with some embodiments of the present disclosure.

(14) FIG. 13 is a block diagram of a system for designing the integrated circuit layout design, in accordance with some embodiments of the present disclosure.

(15) FIG. 14 is a block diagram of an integrated circuit manufacturing system, and an integrated circuit manufacturing flow associated therewith, in accordance with some embodiments.

DETAILED DESCRIPTION

(16) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(17) The terms used in this specification generally have their ordinary meanings in the art and in the specific context where each term is used. The use of examples in this specification, including examples of any terms discussed herein, is illustrative only, and in no way limits the scope and meaning of the disclosure or of any exemplified term. Likewise, the present disclosure is not limited to various embodiments given in this specification.

(18) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(19) As used herein, “around”, “about”, “approximately” or “substantially” shall generally refer to any approximate value of a given value or range, in which it is varied depending on various arts in which it pertains, and the scope of which should be accorded with the broadest interpretation

understood by the person skilled in the art to which it pertains, so as to encompass all such modifications and similar structures. In some embodiments, it shall generally mean within 20 percent, preferably within 10 percent, and more preferably within 5 percent of a given value or range. Numerical quantities given herein are approximate, meaning that the term “around”, “about”, “approximately” or “substantially” can be inferred if not expressly stated, or meaning other approximate values.

(20) Reference is now made to FIG. 1. FIG. 1 is an equivalent circuit of part of an integrated circuit **100**, in accordance with various embodiments. For illustration, the integrated circuit **100** includes transistors **M1** and **M2**. A source of the transistor **M1** is coupled to a power supply terminal **VDD**, and a drain of the transistor **M1** is coupled to an output node **ZN**. A source of the transistor **M2** is coupled to a power supply terminal **VSS**, and a drain of the transistor **M2** is coupled to the output node **ZN**. A gate of the transistor **M1** and a gate of the transistor **M2** are coupled together at an input node **I** of the integrated circuit **100**. In some embodiments, the integrated circuit **100** operates as an inverter. The equivalent circuit of the integrated circuit **100** is given for illustrative purposes. Various configurations of the integrated circuit **100** are within the contemplated scope of the present disclosure. For example, in some embodiments, the integrated circuit **100** is a logic gate circuit including AND, OR, NAND, MUX, Flip-flop, Latch, BUFF or any other types of logic circuit.

(21) In some embodiments, the transistor **M1** is a first conductivity type FET (e.g., P-type), and the transistor **M2** is a second conductivity type FET (e.g., N-type) different from the first conductivity type. The transistors **M1-M2** are given for illustrative purposes. Various configurations of the transistors **M1-M2** are within the contemplated scope of the present disclosure. For example, in some embodiments, the transistor **M1** is an N-type transistor, and the transistor **M2** is a P-type transistor.

(22) Reference is now made to FIG. 2A. FIG. 2A is a perspective diagram of part of the integrated circuit **100** of FIG. 1, in accordance with various embodiments. For illustration, the integrated circuit **100** includes active regions **111-114**, conductive segments **121-124**, gate **130**, power rails **141-142**, metal-zero segments **151-153**, and vias **161-163**. In some embodiments, the power rails **141-142** are disposed in a first layer. The active regions **111-112** and the conductive segments **121-122** are disposed in a second layer above the first layer. The active regions **113-114** and the conductive segments **123-124** are disposed in a third layer above the second layer. The gate **130** extends through the first layer, the second layer, and the third layer. The metal-zero segments **151-153** are disposed in a fourth layer above the third layer.

(23) With reference to FIGS. 1 and 2A, in some embodiments, the active regions **111-112**, the gate **130** and the conductive segments **121-122** are configured with respect to, for example, the transistor **M2** of FIG. 1. The active regions **113-114**, the gate **130** and the conductive segments **123-124** are configured with respect to, for example, the transistor **M1** of FIG. 1. Specifically, the conductive segment **121** corresponds to the drain of the transistor **M2** that is coupled to the output node **ZN**. The conductive segment **122** corresponds to the source of the transistor **M2** that is coupled to the power supply terminal **VSS**. The conductive segment **123** corresponds to the source of the transistor **M1** that is coupled to the power supply terminal **VDD**. The conductive segment **124** corresponds to the drain of the transistor **M1** that is coupled to the output node **ZN**. The gate **130** is configured as the gates of the transistors **M1** and **M2** that are coupled to each other.

(24) With continued reference to FIG. 2A, for illustration, the active regions **111-114** extend in x direction. The active regions **111-112** are separate from the active regions **113-114** in z direction. In some embodiments, the active regions **111-112** are configured to have the second conductivity type (N-type), and the active regions **113-114** are configured to have the first conductivity type (P-type) different from the second conductivity type. Illustratively, the active regions **111-112** are configured as an active area **110A**, and the active regions **113-114** are configured as an active area **110B**. The configurations of FIG. 2A are given for illustrative purposes. Various configurations of the active

regions **111-114** are within the contemplated scope of the present disclosure. For example, in various embodiments, the active regions **111-112** have the P-type conductivity type and the active regions **113-114** have the N-type conductivity type. In alternative embodiments, the active regions **111-114** have the same conductivity type.

(25) For illustration, the conductive segments **121-124** extend in y direction that is different from x and z direction. As shown in FIG. 2A, the conductive segments **121** and **122** are coupled to the active regions **111** and **112** respectively, and the conductive segments **123** and **124** are coupled to the active regions **113** and **114** respectively. In some embodiments, the conductive segments **121-124** are penetrated by the corresponding active regions coupled thereto. In some embodiments, in a plan view, the conductive segment **121** overlaps the conductive segment **123**, and the conductive segment **122** overlaps the conductive segment **124**.

(26) For illustration, as shown in FIG. 2A, the gate **130** extends along y direction and z direction and is disposed between the conductive segments **121**, **123** and conductive segment **122**, **124**. In some embodiments, the gate **130** crosses the active regions **111-114** along y direction.

(27) In some embodiments, the gate **130** includes a gate dielectric layer (not shown) and a gate electrode layer (not shown). In some embodiments, the gate **130** is formed around channel regions of the transistors M1 and M2, in which the channel regions include, for example, structures of round/square wire, nanoslab, nano-sheet, multi-bridge channel, nano-ring or any other suitable kinds of the nano structures.

(28) The power rails **141-142** extend in x direction and are separate from each other along y direction. In some embodiments, the power rail **141** receives the voltage VDD for the integrated circuit **100**, and the power rail **142** receives the voltage VSS for the integrated circuit **100**.

(29) The metal-zero segments **151-153** extend in x direction and are separate from each other along y direction. In some embodiments, the metal-zero segment **151** is coupled to the input node I of FIG. 1 to transmit a control signal to the gate **130** for controlling the transistors M1 and M2. The metal-zero segment **152** is coupled to the output node ZN of FIG. 1 for outputting an output signal from the integrated circuit.

(30) The vias **161-165** extend in z direction. The via **161** is coupled between the conductive segment **123** and the power rail **141**. The via **162** is coupled between the conductive segment **122** and the power rail **142**. The via **163** is coupled between the conductive segment **122** and the conductive segment **124**. The via **164** is coupled between the gate **130** and the metal-zero segment **151**. The via **165** is coupled between the conductive segment **124** and the metal-zero segment **152**.

(31) Reference is now made to FIG. 2B. FIG. 2B is a layout diagram in a plan view of part of the integrated circuit **100** corresponding to a part of FIG. 2A, in accordance with various embodiments. For illustration, the integrated circuit **100** further includes gate stripes **130a-130e**. In some embodiments, at least one in the gate stripes **130a-130e** is configured with respect to, for example, the gate **130**, and the rest of the gate stripes **130a-130e** are referred to as dummy gates, in which in some embodiments, the “dummy” gates are referred to as being not electrically connected as the gates for MOS devices, having no function in the circuit. In some embodiments, the gate stripes **130a-130e** extend in y direction. For simplicity of illustration, the conductive segments **121-124**, the power rails **141-142**, the metal-zero segments **151-153**, and the vias **161-165** are not shown in FIG. 2B.

(32) For illustration, the active areas **110A-110B** of FIG. 2A are configured to have a width W1. As shown in FIG. 2B, the active areas **110A** and **110B** extend in x direction and overlap with each other in the plain view. The configurations of the active areas **110A-110B** are given for illustrative purposes. Various implements of the active areas **110A-110B** are within the contemplated scope of the present disclosure. For example, in some embodiments, the active area **110A** and the active area **110B** have different width from each other.

(33) With continued reference to FIG. 2B, for illustration, the gate stripes **130a-130e** cross the active areas **110A-110B**. In some embodiments, the gate stripes **130a-130e** and the active areas

110A-110B are included in a cell **CELL1** in a layout view. Moreover, in some embodiments, the cell **CELL1** has a cell height **H1** along y direction. The configurations of the cell **CELL1** are given for illustrative purposes. Various implements of the cell **CELL1** are within the contemplated scope of the present disclosure. For example, in some embodiments, instead of multiple gate structures being included in the cell **CELL1**, only one gate structure is included in the cell **CELL1**.

(34) Reference is now made to FIG. 3A. FIG. 3A is a perspective diagram of part of an integrated circuit **300**, in accordance with various embodiments. For illustration, the integrated circuit **300** includes active regions **311a-311c**, **312a-312c**, **313a-313c**, and **314a-314c**, conductive segments **321a-321c**, **322a-322c**, **323a-323c**, and **324a-324c**, gates **330a-330c**, and power rails **341a-341b** and **342a-342b**. For simplicity of illustration, the metal-zero segments **151-153** and the vias **161-165** are not shown in FIG. 3A.

(35) In some embodiments, the active regions **311a-311c**, **312a-312c**, **313a-313c**, and **314a-314c** are configured with respect to, for example, the active regions **111**, **112**, **113** and **114** of FIG. 2A separately. For illustration, the active regions **311a-311c**, **312a-312c**, **313a-313c**, and **314a-314c** are separate from each other in y direction. Illustratively, the active regions **311a-312a** are configured to be as an active area **310A**. The active regions **313a-314a** are configured to be as an active area **310B**. The active regions **311b-312b** are configured to be as an active area **310C**. The active regions **313b-314b** are configured to be as an active area **310D**. The active regions **311c-312c** are configured to be as an active area **310E**. The active regions **313c-314c** are configured to be as an active area **310F**. In some embodiments, the active areas **310A**, **310C**, **310E** are configured with respect to, for example, the active area **110A** of FIG. 2B, and the active areas **310B**, **310D**, **310F** are configured with respect to, for example, the active area **110B** of FIG. 2B.

(36) The conductive segments **321a-321c**, **322a-322c**, **323a-323c**, and **324a-324c** are configured with respect to, for example, the conductive segments **121**, **122**, **123** and **124** of FIG. 2A separately. In some embodiments, as shown in FIG. 3A, the conductive segments **321a-321c**, **322a-322c**, **323a-323c**, and **324a-324c** are separate from each other in y direction.

(37) The gates **330a-330c** are configured with respect to, for example, the gate **130** of FIG. 2A. In some embodiments, as shown in FIG. 3A, the gates **330a-330c** are separate from each other in y direction.

(38) For illustration, as shown in FIG. 3A, the power rails **341a-341b** and **342a-342b** are separate from each other in y direction.

(39) With the configurations of FIG. 3A, in some embodiments, the integrated circuit **300** provides high performance computing ability by implementing multiple active regions operating together corresponding to a source/drain of a transistor. Alternatively stated, one skilled in the art is available to achieve the desired computing ability by implementing the configurations of FIG. 3A.

(40) The configurations of FIG. 3A are given for illustrative purposes. Various implements of the integrated circuit **300** are within the contemplated scope of the present disclosure. For example, in some embodiments, the integrated circuit **300** includes more than three or less than three active regions configured with respect to the active regions **111**, **112**, **113** and/or **114**. In various embodiments, the integrated circuit **300** includes only one pair of power rails instead of having the power rails **341b** and **342b**. Moreover, in some embodiments, the gates **330a-330c** are formed into one gate structure, and/or the conductive segments **321a-321c**, **322a-322c**, **323a-323c**, and/or **324a-324c** are formed into one conductive segment structure.

(41) Reference is now made to FIG. 3B. FIG. 3B is a layout diagram in a plan view of part of the integrated circuit **300** corresponding to a part of FIG. 3A, in accordance with various embodiments. For illustration, the integrated circuit **300** further includes gate stripes **330d-130h**. The gate stripes **330d-130h** are configured with respect to, for example, the gate **330a-330c**. In some embodiments, the gate stripes **330d-330h** extend in y direction. For simplicity of illustration, the conductive segments **321a-321c**, **322a-322c**, **323a-323c**, and **324a-324c**, and the power rails **341a-341b** and **342a-342b** are not shown in FIG. 2B.

(42) For illustration, as shown in FIG. 3B, the active areas **310A-310F**, each having a width of **W1**, extend in x direction. In a layout view, the active areas **310A-310B** overlap with each other, the active areas **310C-310D** overlap with each other, and the active areas **310E-310F** overlap with each other.

(43) In some embodiments, as shown in FIG. 3B, the active areas **310A**, **310C**, and **310E** are close to each other and further operate together with corresponding elements as the transistor **M2** of FIG. 1. Similarly, the active areas **310B**, **310D**, and **310F** are close to each other and further operate together with corresponding elements as the transistor **M1** of FIG. 1.

(44) As discussed above, in some embodiments, the active areas **310A-310F** are arranged in a portion having a width **W2** in the integrated circuit **300** as illustrated in FIG. 3B.

(45) With continued reference to FIG. 3B, for illustration, the gate stripes **330d-330h** cross the active areas **310-310F**. In some embodiments, the gate stripes **330d-330h** and the active areas **310-310F** are included in a cell **CELL2** in a layout view. Moreover, in some embodiments, the cell **CELL2** has a cell height **H2** along y direction. In some embodiments, the cell height **H2** is double of the cell height **H1**. The configurations of the cell **CELL2** are given for illustrative purposes. Various implements of the cell **CELL2** are within the contemplated scope of the present disclosure. For example, in some embodiments, instead of multiple gate structures being included in the cell **CELL2**, only one gate structure is included in the cell **CELL2**.

(46) Reference is now made to FIG. 3C. FIG. 3C is a layout diagram in a plan view of part of the integrated circuit **300** corresponding to a part of FIG. 3A, in accordance with various embodiments. With respect to the embodiments of FIG. 3C, like elements in FIG. 3B are designated with the same reference numbers for ease of understanding. The specific operations of similar elements, which are already discussed in detail in above paragraphs, are omitted herein for the sake of brevity, unless there is a need to introduce the co-operation relationship with the elements shown in FIG. 3C.

(47) Compared with the embodiments of the cell **CELL2** in FIG. 3B, instead of including the active areas **310A-310F**, a cell **CELL3** is formed to include the active areas **310A**, **310B**, **310E** and **310F**. In some embodiments, the cell **CELL3** has the cell height **H2** along y direction.

(48) Reference is now made to FIG. 3D. FIG. 3D is a layout diagram in a plan view of part of the integrated circuit **300** corresponding to a part of FIG. 3A, in accordance with various embodiments. With respect to the embodiments of FIG. 3D, like elements in FIG. 3B are designated with the same reference numbers for ease of understanding. The specific operations of similar elements, which are already discussed in detail in above paragraphs, are omitted herein for the sake of brevity, unless there is a need to introduce the co-operation relationship with the elements shown in FIG. 3D.

(49) Compared with the embodiments of the cell **CELL2** in FIG. 3B, a cell **CELL4** is formed to include active areas **310G**, **310H**, **310I** and **310J**. The active areas **310G** and **310I** are configured with respect to, for example, the active area **310C**. The active areas **310H** and **310J** are configured with respect to, for example, the active area **310D**.

(50) For illustration, the active areas **310G** and **310H** overlap with each other, and the active areas **310I** and **310J** overlap with each other. In some embodiments, the active areas **310G-310J** are disposed between the active area **310A** and **310C**. The configurations of the active areas **310G-310J** in FIG. 3D are given for illustrative purposes. Various implements are within the contemplated scope of the present disclosure. For example, in some embodiments, the active areas **310G-310H** are disposed next to the active areas **310A-310B/310E-310F**, are configured with respect to, for example, the active areas **310A-310B**. In various embodiments, a cell of another design includes more active areas than the cell **CELL4** does for computing.

(51) In some embodiments, as shown in FIG. 3D, the active areas **310A**, **310C**, **310E**, **310G**, and **310I** are close to each other and further operate together with corresponding elements as the transistor **M2** of FIG. 1. Similarly, the active areas **310B**, **310D**, **310F**, **310H**, and **310J** are close to

each other and further operate together with corresponding elements as the transistor M1 of FIG. 1. In some embodiments, the active areas 310A-310H are arranged in a portion having a width W3 larger than the widths W1 and W2.

(52) With the configurations of FIG. 3D, in some embodiments, the cell CELL4 provides higher performance computing ability than the cell CELL1, the cell CELL2 and the cell CELL3 due to more active areas included in the cell for operations.

(53) In some embodiments, the cell CELL4 has a cell height H3 along y direction. In some embodiments, the cell height H3 is triple of the cell height H1. The configurations of the cell CELL4 are given for illustrative purposes. Various implements of the cell CELL4 are within the contemplated scope of the present disclosure. For example, in some embodiments, the cell height H3 is N times of the cell height H1, in which N is an integer greater than 3.

(54) The configurations of FIGS. 3A-3D are given for illustrative purposes. Various implements of FIGS. 3A-3D are within the contemplate scope of the present disclosure. For example, the widths W1, W2, W3 varies with the manufacturing processes of the integrated circuits.

(55) Reference is now made to FIG. 4. FIG. 4 is a layout diagram in a plan view of part of an integrated circuit 400, in accordance with various embodiments. For illustration, the integrated circuit 400 includes cells CELL1a, CELL1b, CELL2a, and CELL3a. In some embodiments, the cells CELL1a-CELL1b are configured with respect to, for example, the cell CELL1 of FIG. 2B. The cell CELL2a is configured with respect to, for example, the cell CELL2 of FIG. 3B. The cell CELL3a is configured with respect to, for example, the cell CELL3 of FIG. 3C.

(56) For illustration, the cell CELL1a abuts the cell CELL1b along x direction. As shown in FIG. 4, top most boundary lines of active areas of the cell CELL1a are aligned with top most boundary lines of active areas of the cell CELL1b along x direction. The cell CELL1a abuts the cell CELL2a along y direction. The cell CELL1b abuts the cell CELL3a along y direction. The cells CELL2a and CELL3a abut each other along x direction. In some embodiments, a top boundary line of at least one active area of the cell CELL2a is aligned with a top boundary line of at least one active area of the cell CELL3a along x direction. For example, top boundary lines of the active areas 310A-310B of the cell CELL2a are aligned with top boundary lines of the active areas 310A-310B of the cell CELL3a. Alternatively stated, one of active areas included in the cell CELL2a and one of active areas included in the cell CELL3a are disposed in the same row in the layout view.

(57) In some embodiments, the alignment configurations of active areas included in multiple abutting cells of FIGS. 4-10 also include, for example, that bottom most boundary lines and/or central lines of active areas of multiple abutting cells align with each other. For the sake of simplicity, the particular portions, of active areas included in multiple abutting cells, referred as aligned with each other are omitted thereafter.

(58) The configurations of FIG. 4 are given for illustrative purposes. Various implements are within the contemplated scope of the present disclosure. For example, in some embodiments, the active areas 310A-310B of the cell CELL2a and the active areas 310A-310B of the cell CELL3a are aligned with each other while active areas 310E-310F of the cell CELL2a and the active areas 310E-310F of the cell CELL3a are not aligned with each other along x direction.

(59) Reference is now made to FIG. 5. FIG. 5 is a layout diagram in a plan view of part of an integrated circuit 500, in accordance with various embodiments. For illustration, the integrated circuit 500 includes cells CELL1c, CELL1d, CELL3b, and CELL4a. In some embodiments, the cells CELL1c-CELL1d are configured with respect to, for example, the cell CELL1 of FIG. 2B. The cell CELL3b is configured with respect to, for example, the cell CELL3 of FIG. 3C. The cell CELL4a is configured with respect to, for example, the cell CELL4 of FIG. 3D.

(60) For illustration, the cell CELL1c abuts the cell CELL1d along y direction. The cells CELL1c and CELL1d abut on one side the cell CELL3b and abut the cell CELL4a on an opposite side, along x direction. In some embodiments, active areas 110A-110B of the cell CELL1c are aligned with active areas 310A-310B of the cell CELL3b and active areas 310H-310J of the cell CELL4a

along x direction. Active areas **110A-110B** of the cell **CELL1d** are aligned with active areas **310E-310F** of the cell **CELL3b** and active areas **310E-310F** of the cell **CELL4a** along x direction. In some embodiments, the active area **110A/110B** of the cell **CELL1c/CELL1d** is aligned with one of active areas of the cell **CELL4a** and/or one of the cell **CELL3b** along x direction. Alternatively stated, in some embodiments, one of active areas included in the cell **CELL1c** and/or in the **CELL1d**, one of active areas included in the cell **CELL3b**, and one of active areas included in the cell **CELL4a** are disposed in the same row in the layout view.

(61) The configurations of FIG. 5 are given for illustrative purposes. Various implements are within the contemplated scope of the present disclosure. For example, in some embodiments, the cell **CELL3b** abuts the cell **CELL1c** and **CELL1b** on a side and abuts the cell **CELL4a** on an opposite side.

(62) Reference is now made to FIG. 6. FIG. 6 is a layout diagram in a plan view of part of an integrated circuit **600**, in accordance with various embodiments. With respect to the embodiments of FIG. 6, like elements in FIGS. 4 and 5 are designated with the same reference numbers for ease of understanding. The specific operations of similar elements, which are already discussed in detail in above paragraphs, are omitted herein for the sake of brevity, unless there is a need to introduce the co-operation relationship with the elements shown in FIG. 6.

(63) Compared with the embodiments of the integrated circuit **500** of FIG. 5, for illustration, the integrated circuit **600** further includes the cells **CELL1a**, **CELL2a**, cell **CELL3c** and excludes the cell **CELL3b**. For illustration, the cell **CELL2a** is disposed along y direction between the cell **CELL1a** and the adjacent cells **CELL1c** and **CELL1d**. The cell **CELL3c** abuts the cell **CELL1a** along x direction and abuts the cell **CELL4a** along y direction. The cell **CELL4a** abuts the cells **CELL1c**, **CELL1d** and **CELL2a** along x direction.

(64) As shown in FIG. 6, active areas **310A-310B** of the cell **CELL3c** are aligned to the active areas **110A-110B** of the cell **CELL1a**, and active areas **310E-310F** of the cell **CELL3c** are aligned to the active areas **310A-310B** of the cell **CELL2a** along x direction. The active areas **310E-310F** of the cell **CELL2a** are aligned to the active areas **310A-310B** of the cell **CELL4a** along x direction. Alternatively stated, the active areas **310E-310F** of the cell **CELL3c** are disposed directly next to the active areas **310A-310B** of the cell **CELL4a** along y direction in the layout view. The active areas **110A-110B** of the cell **CELL1a** are disposed directly next to the active areas **310A-310B** of the cell **CELL2a** along y direction in the layout view. The active areas **310E-310F** of the cell **CELL2a** are disposed directly next to the active areas **110A-110B** of the cell **CELL1c** along y direction in the layout view.

(65) The configurations of FIG. 6 are given for illustrative purposes. Various implements are within the contemplated scope of the present disclosure. For example, in some embodiments, the active areas **310A-310B** of the cell **CELL3c** and the active areas **110A-110B** of the cell **CELL1a** are aligned with each other while active areas **310E-310F** of the cell **CELL3c** and the active areas **310A-310B** of the cell **CELL2a** are not aligned with each other along x direction.

(66) Reference is now made to FIG. 7. FIG. 7 is a layout diagram in a plan view of part of an integrated circuit **700**, in accordance with various embodiments. With respect to the embodiments of FIG. 7, like elements in FIGS. 4-6 are designated with the same reference numbers for ease of understanding. The specific operations of similar elements, which are already discussed in detail in above paragraphs, are omitted herein for the sake of brevity, unless there is a need to introduce the co-operation relationship with the elements shown in FIG. 7.

(67) Compared with the embodiments of the integrated circuit **600** in FIG. 6, the integrated circuit **700** further includes the cells **CELL1b**, **CELL3a**, and **CELL3b**. With reference as discussed above, the cell **CELL1b** and the cell **CELL3a** abut the cell **CELL1a** and the cell **CELL2a** respectively. Furthermore, the cell **CELL3b** abuts the cell **CELL2a** along y direction and abuts the cell **CELL1c** and the cell **CELL1d** along x direction.

(68) As shown in FIG. 7, the active areas **310A-310B** of the cell **CELL3c** are aligned with the

active areas **110A-110B** of the cell **CELL1a** and the active areas **110A-110B** of the cell **CELL1b**. The active areas **310E-310F** of the cell **CELL3c** are aligned with the active areas **310A-310B** of the cell **CELL2a** and the active areas **310A-310B** of the cell **CELL3a**. The active areas **310E-310F** of the cell **CELL2a** are aligned with the active areas **310E-310F** of the cell **CELL3a** and the active areas **310A-310B** of the cell **CELL4a**.

(69) In some approaches, active areas of a first conductivity type and active areas of a second conductivity type in cells are arranged in the same layer and separate from each other in y direction. Active areas of the same conductivity type in two adjacent cells of the cells are aligned in x direction. Moreover, along two opposite direction along y direction, cells of high computing performance have a half single cell height difference from cells of normal performance. Because of the cell height difference, when the cells of normal performance abut the cells of high computing performance along y direction, white spaces of half single cell height between the cells of normal performance and the cells of high computing performance are produced in the layout view. Accordingly, in such approaches, the layout of the integrated circuit suffers the area penalty due to the white spaces.

(70) Compared with aforementioned approaches, with the configurations of the present disclosure, no white spaces are induced when cells abut with each other along y direction. Accordingly, the greater cell density and better layout flexibility are provided in the present disclosure.

(71) The configurations of FIGS. 4-7 are given for illustrative purposes. Various implements are within the contemplated scope of the present disclosure. For example, in some embodiments, the cells **CELL3a-CELL3c** are replaced by the cell **CELL2a** in order to gain high computing performance.

(72) Reference is now made to FIG. 8. FIG. 8 is a layout diagram in a plan view of part of an integrated circuit **800**, in accordance with various embodiments. For illustration, the integrated circuit **800** includes cells **CELL1e**, **CELL3d**, and **CELL5**. In some The cell **CELL1e** is configured with respect to, for example, the cell **CELL1** of FIG. 2B. The cell **CELL3d** is configured with respect to, for example, the cell **CELL3** of FIG. 3C.

(73) As shown in FIG. 8, the cell **CELL5** includes multiple first active areas of the first conductivity type (e.g., N-type) and multiple second active areas of the second conductivity type (e.g., P-type). The first active areas and the second active areas overlap with each other in the layout view. In some embodiments, the first active areas are configured with respect to, for example, the active areas **310A**, **310C**, and **310E** of FIG. 3B. The second active areas are configured with respect to, for example, the active areas **310B**, **310D**, and **310F** of FIG. 3B.

(74) In some embodiments, the first active areas of the cell **CELL5** are close to each other and further operate together with corresponding elements as the transistor **M2** of FIG. 1. Similarly, the second active areas of the cell **CELL5** are close to each other and further operate together with corresponding elements as the transistor **M1** of FIG. 1. In some embodiments, the first active areas and the second active areas of the cell **CELL5** are arranged in a portion having width **W4** larger than the widths **W1-W3** in the integrated circuit **800**.

(75) For illustration, the cell **CELL5** has a cell height **H4** along y direction. In some embodiments, the cell height **H4** is five times larger than the cell height **H1**. The configurations of the cell **CELL5** are given for illustrative purposes. Various implements of the cell **CELL5** are within the contemplated scope of the present disclosure. For example, in some embodiments, the cell height **H4** is a total of cell height of the cells abutting the cell **CELL5**.

(76) With the configurations of the cell **CELL5** in FIG. 8, in some embodiments, the cell **CELL5** provides higher performance computing ability than the cells **CELL1-CELL4** due to more active areas included in the cell for operations.

(77) With continued reference to FIG. 8. The cell **CELL1e** abuts the cell **CELL5** along x direction and abuts the cell **CELL3d** along y direction. The cell **CELL3d** abuts the cell **CELL5** along x direction. In some embodiments, active areas **110A-110B** of the cell **CELL1e** are aligned with one

active area of the first active areas and one active area of the second active areas of the cell CELL5. Active areas **310A-310B** and **310E-310F** of the cell CELL3d are aligned with two active areas of the first active areas and two active areas of the second active areas of the cell CELL5.

(78) Reference is now made to FIG. 9. FIG. 9 is a layout diagram in a plan view of part of an integrated circuit **900**, in accordance with various embodiments. With respect to the embodiments of FIG. 9, like elements in FIG. 8 are designated with the same reference numbers for ease of understanding. The specific operations of similar elements, which are already discussed in detail in above paragraphs, are omitted herein for the sake of brevity, unless there is a need to introduce the co-operation relationship with the elements shown in FIG. 9.

(79) Compared with the embodiments of the integrated circuit **800** of FIG. 8, for illustration, the integrated circuit **900** further includes a cell CELL3e and excludes the cell CELL3d. In some embodiments, the cell CELL3e is configured with respect to, for example, the cell CELL3 of FIG. 3C.

(80) For illustration, the cell CELL3e abuts the cell CELL5 along x direction. In some embodiments, active areas **310A-310B** of the cell CELL3e are aligned with the one active area of the first active areas, the one active area of the second active areas of the cell CELL5, and the active areas **110A-110B** of the cell CELL1e. Moreover, active areas **310E-310F** of the cell CELL3e are aligned with another one active area of the first active areas and another one active area of the second active areas of the cell CELL5.

(81) Reference is now made to FIG. 10. FIG. 10 is a layout diagram in a plan view of part of an integrated circuit **1000**, in accordance with various embodiments. With respect to the embodiments of FIG. 10, like elements in FIGS. 8 and 9 are designated with the same reference numbers for ease of understanding. The specific operations of similar elements, which are already discussed in detail in above paragraphs, are omitted herein for the sake of brevity, unless there is a need to introduce the co-operation relationship with the elements shown in FIG. 10.

(82) Compared with the embodiments of the integrated circuit **900** in FIG. 9, the integrated circuit **1000** further includes the cells CELL3d, cells CELL1f, CELL3f, and CELL3g. In some embodiments, the cell CELL1f is configured with respect to, for example, the cell CELL1 of FIG. 2B. The cells CELL3f and CELL3g are configured with respect to, for example, the cell CELL3 of FIG. 3C.

(83) For illustration, the cell CELL3d abuts the cell CELL5 along x direction and the cell CELL3f along y direction. The cell CELL3f abuts the cell CELL5 along x direction. The cell CELL1f is disposed between the cell CELL3e and the cell CELL3g and abuts the cell CELL5 along x direction. The cell CELL3g abuts the cell CELL5 along the y direction and the cell CELL5 along x direction.

(84) As shown in FIG. 10, the active areas **310E-310F** of the cell CELL3e are aligned the another one active area of the first active areas, the another one active area of the second active areas of the cell CELL5, and active areas **310A-310B** of the cell CELL3d. Active areas **110A-110B** of the cell CELL1f are aligned with still another one of the first active areas of the CELL5, still another one of the second active areas of the CELL5, and active areas **310E-310F** of the cell CELL3d. Moreover, active areas **310A-310B** are aligned with still another active areas of the first active areas of the CELL5, still another one of the second active areas of the CELL5, and active areas **310A-310B** of the cell CELL3f. Active areas **310E-310F** are aligned with still another active areas of the first active areas of the CELL5, still another one of the second active areas of the CELL5, and active areas **310E-310F** of the cell CELL3f. Alternatively stated, a first portion of the first active areas of the cell CELL5 and a first portion of the second active areas of the cell CELL5 are aligned with active areas included in the cells abutting the cell CELL5. A second portion of the first active areas of the cell CELL5 and a second portion of the second active areas of the cell CELL5 are not aligned with active areas included in the cells abutting the cell CELL5.

(85) In some embodiments, a total number of the first active areas and the second active areas of

the cell CELL5 is greater than a total number of active areas included in the cells CELL1e, CELL3d, and CELL3f. Similarly, the total number of the first active areas and the second active areas of the cell CELL5 is greater than a total number of active areas included in the cells CELL3e, CELL1f, and CELL3g.

(86) In some approaches, for example, high computing performance cells include, in a layer, active areas of a first conductivity type and active areas of a second conductivity type arranged next to the active areas of the first conductivity type. When two high computing performance cells abut each other along y direction, there is no space for extra active areas being arranged between the active areas of the first conductivity type and the active areas of the second conductivity type.

(87) In contrast to such approaches mentioned above, with the configuration of the present disclosure, when two high computing performance cells abut each other along y direction and have a same total cell height as aforementioned approaches, extra active areas of the first conductivity type can be arranged next to the active areas of the first conductivity type of the two high computing performance cells in the first layer, and extra active areas of the second conductivity type can be arranged next to the active areas of the second conductivity type of the two high computing performance cells in the second layer. Alternatively stated, more active areas for operation can be added to integrated circuits, compared to the some approaches. Accordingly, the present disclosure provides better flexibility of layout and performance of cells in integrated circuits.

(88) The configurations of FIGS. 8-10 are given for illustrative purposes. Various implements are within the contemplated scope of the present disclosure. For example, in some embodiments, the numbers of the first active areas and the second active areas of the cell CELL5 are different from each other.

(89) Reference is now made to FIG. 11. FIG. 11 is a flow chart of a method 1100 of generating a layout design for fabricating the integrated circuit 100, 300, 400, 500, 600, 700, 800, 900, or 1000, in accordance with some embodiments of the present disclosure. It is understood that additional operations can be provided before, during, and after the processes shown by FIG. 11, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method 1100. The method 1100 includes operations S1110-S1130 that are described below with reference to FIG. 7.

(90) In operation 1110, generating the cell CELL1a with the cell height H1 is performed. In some embodiments, the cell CELL1a includes the active area 110A and the active area 110B overlapping the active area 110A, as shown in FIG. 2B.

(91) In operation 1120, generating the cell CELL3c with the cell height H2 is performed. In some embodiments, the cell CELL3c includes the active areas 310A-310B and the active areas 310E-310F overlapping the active areas 310A-310B.

(92) In operation 1130, arranging the cell CELL1a abutting the cell CELL3c is performed. In some embodiments, the active area 110A and one of the active areas 310A-310B extend in a first row, and/or the active area 110B and one of the active areas 310A-310B extend in the first row.

(93) In some embodiments, the method 1100 further includes operations of generating the cell CELL2a abutting the cell CELL1a and the cell CELL3c. Specifically, the cell CELL3c includes the active areas 310A, 310C and 310E and the active areas 310B, 310D, and 310F overlapping the active areas 310A, 310C and 310E, separately. In some embodiments, the number of the active areas 310A, 310C and 310E of the cell CELL2a is greater than the number of the active areas 310A and 310E of the cell CELL3c. Moreover, one of the active areas 310A, 310C and 310E of the cell CELL2a and one of the active areas 310A and 310E extend in a second row different from the first row. In various embodiments, the cell CELL2a and the cell CELL3c have the same cell height, for example, the cell height H2.

(94) In some embodiments, the method 1100 further includes operations of generating the cell CELL4a, in which the cell CELL4a includes the active areas 310A, 310C, 310E, 310G, and 310I

and the active areas **310B**, **310D**, **310F**, **310H**, and **310J** overlapping the active areas **310A**, **310C**, **310E**, **310G**, and **310I**. The method **1100** further includes operations of arranging the cell **CELL4a** abuts the cells **CELL1a**, **CELL2a**, and **CELL3c**. In some embodiments, the cell **CELL4a** includes a greater number of active areas than that included in the cell **CELL2a**, and the cell **CELL2a** includes a greater number of active areas than that included in the cell **CELL3c**.

(95) In some embodiments, the method **1100** further includes operations of manufacturing at least one element of the integrated circuit **100**, **300**, **400**, **500**, **600**, **700**, **800**, **900**, or **1000**, based on the layout generated through the operations mentioned above.

(96) Reference is now made to FIG. **12**. FIG. **12** is a flow chart of a method **1200** of fabricating the integrated circuit **100**, **300**, **400**, **500**, **600**, **700**, **800**, **900**, or **1000** based on the layout design generated by the method **1100** of FIG. **11**, in accordance with some embodiments of the present disclosure. It is understood that additional operations can be provided before, during, and after the processes shown by FIG. **11**, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method **1200**. The method **1200** includes operations **S1210-S1250** that are described below with reference to FIGS. **3A** and **4**.

(97) In operation **1210**, the power rails **341a-342a** and **341c-342c** of the cells **CELL1a**, **CELL1b**, **CELL2a**, and **CELL3a** are formed and disposed in a first layer.

(98) In operation **1220**, the gates **330a-330c** are formed through the first layer, the second layer above the first layer, and the third layer above the second layer. In some embodiments, each one of the gates **330a-330c** includes an interfacial layer (not shown) and a polysilicon (or poly) layer (not shown) over the interfacial layer. In some embodiments, the gates **330a-330c** further include a gate dielectric layer (not shown) and a metal gate layer (not shown) disposed between the interfacial layer and the poly layer. In some embodiments, the gates **330a-330c** includes one or more metal layers in place of the poly layer. In various embodiments, the interfacial layer includes a dielectric material including, for example, silicon oxide (SiO_2) or silicon oxynitride (SiON), and is able to be formed by chemical oxidation, thermal oxidation, atomic layer deposition (ALD), chemical vapor deposition (CVD), and/or other suitable methods. In some embodiments, the polysilicon layer is formed by suitable deposition processes including, for example, low-pressure chemical vapor deposition (LPCVD) and plasma-enhanced CVD (PECVD). In some embodiments, the gate dielectric layer uses a high-k dielectric material including, for example, hafnium oxide (HfO_2), Al_2O_3 , lanthanide oxides, TiO_2 , HfZrO , Ta_2O_3 , HfSiO_4 , ZrO_2 , ZrSiO_2 , combinations thereof, or other suitable material, and the gate dielectric layer is formed by ALD and/or other suitable methods. The metal gate layer includes a p-type work function metal or an n-type work function metal, and is deposited by CVD, PVD, and/or other suitable process. Exemplary p-type work function metals include TiN , TaN , Ru , Mo , Al , WN , ZrSi_2 , MoSi_2 , TaSi_2 , NiSi_2 , WN , other suitable p-type work function materials, or combinations thereof. Exemplary n-type work function metals include Ti , Ag , TaAl , TaAlC , TiAlN , TaC , TaCN , TaSiN , Mn , Zr , other suitable n-type work function materials, or combinations thereof. The one or more metal layers use aluminum (Al), tungsten (W), copper (Cu), cobalt (Co), and/or other suitable materials; and are formed by CVD, PVD, plating, and/or other suitable processes. The formations and/or materials associated with the gates **330a-330c** are given for illustrative purposes. Various formations and/or materials associated with the gates **330a-330c** are within the contemplated scope of the present disclosure.

(99) In some embodiments, the gate **330a-330c** are implemented by being included in multilayer stacks, for example, including structures of round/square wire, nanoslab, nano-sheet, multi-bridge channel, nano-ring or any other suitable kinds of the nano structures.

(100) In operation **1230**, the active regions **311a-311c** and the conductive segments **321a-321c**, **322a-322c** of the cells of FIG. **4** are formed in the second layer. In some embodiments, the conductive segments **321a-321c**, **322a-322c** are formed after the formation of the active regions **311a-311c**.

(101) In some embodiments, after the formation of the active regions **311a-311c** and the conductive

segments **321a-321c**, **322a-322c**, an isolation layer is formed above the formation of the active regions **311a-311c** and the conductive segments **321a-321c**, **322a-322c**, in order to electrically isolate the conductive elements in the second layer and the following third layer.

(102) In operation **1240**, the active regions **311a-311c** and the conductive segments **321a-321c**, **322a-322c** of the cells of FIG. **4** are formed in the second layer. In some embodiments, the conductive segments **321a-321c**, **322a-322c** are formed after the formation of the active regions **311a-311c**.

(103) As discussed above, in some embodiments, the active areas of a high performance cell (i.e., the cell **CELL2a** of FIG. **4**) and the active areas of a normal cell (i.e., the cells **CELL1a-CELL1b**, and **CELL3a**) are formed and/or patterned in the same procedure.

(104) In operation **1250**, the metal-zero segments **151-153** are formed. In some embodiments, metal-one segments arranged above the metal-zero segments are formed. The arrangements of metal-zero segments, metal-one segments and/or more metal layers are on the basis of actual metal routing of the integrated circuit.

(105) In some embodiments, the method **1200** also includes forming vias between conductive elements (i.e., the via **161** coupled between the power rail **141** and the conductive segment **123**, etc.) for implementing the integrated circuit.

(106) FIG. **13** is a block diagram of a system for designing the integrated circuit layout design, in accordance with some embodiments of the present disclosure. One or more operations of the method **1100** as discussed above with respect to FIG. **11** are implementable using the IC device design system **1300**, in accordance with some embodiments.

(107) In some embodiments, IC device design system **1300** is a computing device including a hardware processor **1302** and a non-transitory computer-readable storage medium **1304**. Non-transitory computer-readable storage medium **1304**, amongst other things, is encoded with, i.e., stores, computer program codes, i.e., a set of executable instructions **1306**. Execution of instructions **1306** by the hardware processor **1302** represents (at least in part) an IC device design system which implements a portion or all of, e.g., the method **1100** discussed above with respect to FIG. **11**. (hereinafter, the noted processes and/or methods).

(108) The processor **1302** is electrically coupled to non-transitory computer-readable storage medium **1304** via a bus **1308**. The processor **1302** is also electrically coupled to an I/O interface **1310** and a fabrication tool **1316** by bus **1308**. A network interface **1313** is also electrically connected to processor **1302** via bus **1308**. Network interface **1313** is connected to a network **1314**, so that processor **1302** and non-transitory, computer-readable storage medium **1304** are capable of being connected to external elements via network **1314**. Processor **1302** is configured to execute the instructions **1306** encoded in non-transitory computer-readable storage medium **1304** in order to cause IC device design system **1300** to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, processor **1302** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

(109) In one or more embodiments, non-transitory computer-readable storage medium **1304** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, non-transitory computer-readable storage medium **1304** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In one or more embodiments using optical disks, non-transitory computer-readable storage medium **1304** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

(110) In one or more embodiments, non-transitory computer-readable storage medium **1304** stores the instructions **1306** configured to cause IC device design system **1300** to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, non-transitory

computer-readable storage medium **1304** also stores information which facilitates performing a portion or all of the noted processes and/or methods. In various embodiments, non-transitory computer-readable storage medium **1304** stores one or a combination of at least one IC layout designs **1320** or at least one design specification **1322**, each discussed above with respect to FIGS. **2B**, **3B-10** and the method **1100** in FIG. **11**.

(111) IC device design system **1300** includes I/O interface **1310**. I/O interface **1310** is coupled to external circuitry. In various embodiments, I/O interface **1310** includes one or a combination of a keyboard, keypad, mouse, trackball, trackpad, display, touchscreen, and/or cursor direction keys for communicating information and commands to and/or from processor **1302**.

(112) IC device design system **1300** also includes network interface **1313** coupled to processor **1302**. Network interface **1313** allows IC device design system **1300** to communicate with network **1314**, to which one or more other computer systems are connected. Network interface **1313** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interfaces such as ETHERNET, USB, or IEEE-1364. In one or more embodiments, a portion or all of the noted processes and/or methods, is implemented in two or more systems **1300**.

(113) IC device design system **1300** also includes the fabrication tool **1316** coupled to processor **1302**. The fabrication tool **1316** is configured to fabricate integrated circuits, e.g., the integrated circuit **100** illustrated in FIG. **1**, according to the design files processed by the processor **1302**.

(114) IC device design system **1300** is configured to receive information through I/O interface **1310**. The information received through I/O interface **1310** includes one or a combination of at least one design rule instructions, at least one set of criteria, at least one design rule, at least one DRM, and/or other parameters for processing by processor **1302**. The information is transferred to processor **1302** via bus **1308**. IC device design system **1300** is configured to transmit and/or receive information related to a user interface through I/O interface **1310**.

(115) In some embodiments, a portion or all of the noted processes and/or methods is implemented as a standalone software application for execution by a processor. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is a part of an additional software application. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a plug-in to a software application. In some embodiments, at least one of the noted processes and/or methods is implemented as a software application that is a portion of an EDA tool. In some embodiments, an IC layout diagram is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool.

(116) In some embodiments, the processes are realized as functions of a program stored in a non-transitory computer-readable recording medium. Examples of a non-transitory computer-readable recording medium include, but are not limited to, external/removable and/or internal/built-in storage or memory unit, e.g., one or more of an optical disk, such as a DVD, a magnetic disk, such as a hard disk, a semiconductor memory, such as a ROM, a RAM, a memory card, and the like.

(117) By being usable to implement one or more operations of the method **1100** in FIG. **11**, IC device design system **1300** and a non-transitory computer-readable storage medium, e.g., non-transitory computer-readable storage medium **1304**, enable the benefits discussed above with respect to the method **1100** in FIG. **11**.

(118) FIG. **14** is a block diagram of IC manufacturing system **1400**, and an IC manufacturing flow associated therewith, in accordance with some embodiments of the present disclosure. In some embodiments, based on a layout design, at least one of (A) one or more semiconductor masks or (B) at least one component in a layer of a semiconductor integrated circuit is fabricated using the IC manufacturing system **1400**.

(119) In FIG. **14**, the IC manufacturing system **1400** includes entities, such as a design house **1420**, a mask house **1430**, and an IC manufacturer/fabricator (“fab”) **1450**, that interact with one another

in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **1460**. The entities in system **1400** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house **1420**, mask house **1430**, and IC fab **1450** is owned by a single larger company. In some embodiments, two or more of design house **1420**, mask house **1430**, and IC fab **1450** coexist in a common facility and use common resources.

(120) Design house (or design team) **1420** generates an IC design layout diagram (or design) **1422** based on the method **1100** in FIG. **11**, discussed above with respect to FIGS. **2B**, **3B-10**. IC design layout diagram **1422** includes various geometrical patterns that correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **1460** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout diagram **1422** includes various IC features, such as an active area, gate electrode, source and drain, metal lines or vias of an interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **1420** implements a proper design procedure including the method **1100** in FIG. **11**, discussed above with respect to FIGS. **2B**, **3B-10**, to form IC design layout diagram **1422**. The design procedure includes one or more of logic design, physical design or place and route. IC design layout diagram **1422** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout diagram **1422** can be expressed in a GDSII file format or DFII file format.

(121) Mask house **1430** includes data preparation **1432** and mask fabrication **1444**. Mask house **1430** uses IC design layout diagram **1422** to manufacture one or more masks **1445** to be used for fabricating the various layers of IC device **1460** according to IC design layout diagram **1422**. Mask house **1430** performs mask data preparation **1432**, where IC design layout diagram **1422** is translated into a representative data file (“RDF”). Mask data preparation **1432** provides the RDF to mask fabrication **1444**. Mask fabrication **1444** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) **1445** or a semiconductor wafer **1453**. The design layout diagram **1422** is manipulated by mask data preparation **1432** to comply with particular characteristics of the mask writer and/or requirements of IC fab **1450**. In FIG. **14**, mask data preparation **1432** and mask fabrication **1444** are illustrated as separate elements. In some embodiments, mask data preparation **1432** and mask fabrication **1444** are collectively referred to as mask data preparation.

(122) In some embodiments, mask data preparation **1432** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout diagram **1422**. In some embodiments, mask data preparation **1432** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

(123) In some embodiments, mask data preparation **1432** includes a mask rule checker (MRC) that checks the IC design layout diagram **1422** that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some embodiments, the MRC modifies the IC design layout diagram **1422** to compensate for limitations during mask fabrication **1444**, which may undo part of the modifications performed by OPC in

order to meet mask creation rules.

(124) In some embodiments, mask data preparation **1432** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **1450** to fabricate IC device **1460**. LPC simulates this processing based on IC design layout diagram **1422** to create a simulated manufactured device, such as IC device **1460**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus (“DOF”), mask error enhancement factor (“MEEF”), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout diagram **1422**.

(125) It should be understood that the above description of mask data preparation **1432** has been simplified for the purposes of clarity. In some embodiments, data preparation **1432** includes additional features such as a logic operation (LOP) to modify the IC design layout diagram **1422** according to manufacturing rules. Additionally, the processes applied to IC design layout diagram **1422** during data preparation **1432** may be executed in a variety of different orders.

(126) After mask data preparation **1432** and during mask fabrication **1444**, a mask **1445** or a group of masks **1445** are fabricated based on the modified IC design layout diagram **1422**. In some embodiments, mask fabrication **1444** includes performing one or more lithographic exposures based on IC design layout diagram **1422**. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) **1445** based on the modified IC design layout diagram **1422**. Mask **1445** can be formed in various technologies. In some embodiments, mask **1445** is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask version of mask **1445** includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the binary mask. In another example, mask **1445** is formed using a phase shift technology. In a phase shift mask (PSM) version of mask **1445**, various features in the pattern formed on the phase shift mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **1444** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in semiconductor wafer **1453**, in an etching process to form various etching regions in semiconductor wafer **1453**, and/or in other suitable processes.

(127) IC fab **1450** includes wafer fabrication **1452**. IC fab **1450** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC Fab **1450** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business.

(128) IC fab **1450** uses mask(s) **1445** fabricated by mask house **1430** to fabricate IC device **1460**. Thus, IC fab **1450** at least indirectly uses IC design layout diagram **1422** to fabricate IC device **1460**. In some embodiments, semiconductor wafer **1453** is fabricated by IC fab **1450** using mask(s) **1445** to form IC device **1460**. In some embodiments, the IC fabrication includes performing one or more lithographic exposures based at least indirectly on IC design layout diagram **1422**.

Semiconductor wafer **1453** includes a silicon substrate or other proper substrate having material

layers formed thereon. Semiconductor wafer 1453 further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

(129) As described above, the integrated circuit in the present disclosure includes cells of different cell heights. By implementing the complementary field-effect transistor structure, the white spaces between normal cells and high performance cells abutting the normal cells are eliminated.

Accordingly, the present disclosure provides the integrated circuit with high cell density and good layout flexibility.

(130) In some embodiments, an integrated circuit is disclosed, including a first transistor of a first conductivity type including a first active area extending in a first direction; a second transistor of the first conductivity type including at least two second active areas extending in the first direction and a first gate stripe crossing the at least two second active areas; and a third transistor of a second conductivity type that is stacked on the second transistor and includes at least two third active areas arranged above the at least two second active areas. A top most boundary line of the first active area is aligned with a top most boundary line of one of the at least two third active areas in a layout view. In some embodiments, the first conductivity type is N type, and the second conductivity type is P type. In some embodiments, the first transistor further includes a fourth active area separate from the first active area in a second direction different from the first direction. In some embodiments, a cell height of a first cell including the first transistor is equal to a cell height of a second cell including the second and third transistors. In some embodiments, a bottom most boundary line of the fourth active area is aligned with a bottom most boundary line of one of the at least two third active areas in the layout view. In some embodiments, a cell height of a first cell including the first transistor is different from a cell height of a second cell including the second and third transistors. In some embodiments, the integrated circuit further includes a fourth transistor of the second conductivity type that is stacked under the first transistor and includes a fourth active area extending in the first direction and a second gate stripe extending in a second direction different from the first direction to pass across the fourth active area. In some embodiments, the integrated circuit further includes fifth and sixth transistors of different conductivity types in a first cell. The fifth and sixth transistors include multiple fifth active areas overlapping each other in the layout view. The second and third transistors are included in a second cell abutting the first cell in the second direction. The first gate stripe further crosses at least one in the fifth active areas.

(131) Also disclosed is an integrated circuit that includes a first cell having a first cell height and including a first active area in a first layer and a second cell having a second cell height different from the first cell height. The second cell includes a first plurality of active areas in the first layer. The integrated circuit further includes a third cell having a third cell height greater than the first cell height and the second cell height. The third cell includes a second plurality of active areas in a second layer below the first layer. The first and second cells abut the third cell in a first direction, and the first and second cells abut each other in a second direction. The first active area and a first area in the second plurality of active areas align with each other along the first direction, and first and second areas in the first plurality of active areas align with second and third areas in the second plurality of active areas along the first direction. In some embodiments, the first cell height is smaller than the second cell height and the second cell height. In some embodiments, a number of the first plurality of active areas is smaller than a number of the second plurality of active areas. In some embodiments, the second cell further includes a third plurality of active areas in the second layer. The third plurality of active areas overlap the first plurality of active areas, and align with corresponding ones of the second plurality of active areas in a layout view. In some embodiments, the second cell further includes a non-doped region between the first and second areas in the first plurality of active areas, and a third area in the second plurality of active areas is disposed between the second and third areas in the second plurality of active areas. In some embodiments, the first active area align with a fourth area in the second plurality of active areas along the first direction.

In some embodiments, the integrated circuit further includes multiple gate stripes that extend continuously in the second direction and shared by the first and second cells.

(132) Also disclosed is a method that includes the operation below: forming a first plurality of active areas that extend in a first row, wherein one of the first plurality of active areas overlaps another one of the first plurality of active areas in a layout view; forming a second plurality of active areas that extend in a second row, wherein one of the second plurality of active areas overlaps another one of the second plurality of active areas in the layout view; forming a third plurality of active areas that extend in a third row between the first and second rows; and forming multiple gate structures extending to cross at least one of the first plurality of active areas, at least one of the second plurality of active areas, and at least one of the third plurality of active areas. In some embodiments, two overlapping active areas have different conductivity types. In some embodiments, the third plurality of active areas are shorter than the first plurality of active areas and the second plurality of active areas. In some embodiments, one of the third plurality of active areas overlaps another one of the third plurality of active areas in the layout view. In some embodiments, the method further includes forming a fourth plurality of active areas that extend in a fourth row adjacent to the second row, wherein one of the fourth plurality of active areas overlaps another one of the fourth plurality of active areas in the layout view. The third and fourth plurality of active areas are shorter than the first plurality of active areas and the second plurality of active areas.

(133) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated circuit, comprising: a first transistor of a first conductivity type comprising a first active area extending in a first direction; a second transistor of the first conductivity type comprising at least two second active areas extending in the first direction and a first gate stripe crossing the at least two second active areas; and a third transistor of a second conductivity type that is stacked on the second transistor and comprises at least two third active areas arranged above the at least two second active areas, wherein a top most boundary line of the first active area is aligned with a top most boundary line of one of the at least two third active areas in a layout view, wherein the third transistor is separated from the first transistor and the second transistor along the first direction in the layout view.
2. The integrated circuit of claim 1, wherein the first conductivity type is N type, and the second conductivity type is P type.
3. The integrated circuit of claim 1, wherein the first transistor further comprises a fourth active area separate from the first active area in a second direction different from the first direction.
4. The integrated circuit of claim 3, wherein a cell height of a first cell including the first transistor is equal to a cell height of a second cell including the second and third transistors.
5. The integrated circuit of claim 3, wherein a bottom most boundary line of the fourth active area is aligned with a bottom most boundary line of one of the at least two third active areas in the layout view.
6. The integrated circuit of claim 1, wherein a cell height of a first cell including the first transistor is different from a cell height of a second cell including the second and third transistors.

7. The integrated circuit of claim 1, further comprising: a fourth transistor of the second conductivity type that is stacked on the first transistor and comprises a fourth active area extending in the first direction and a second gate stripe extending in a second direction different from the first direction to pass across the fourth active area.
8. The integrated circuit of claim 7, further comprising: fifth and sixth transistors of different conductivity types in a first cell, wherein the fifth and sixth transistors comprise a plurality of fifth active areas overlapping each other in the layout view, wherein the second and third transistors are included in a second cell abutting the first cell in the second direction, wherein the first gate stripe further crosses at least one in the plurality of fifth active areas.
9. An integrated circuit, comprising: a first cell having a first cell height and comprising a first active area in a first layer; a second cell having a second cell height different from the first cell height, wherein the second cell comprises a first plurality of active areas in the first layer; and a third cell having a third cell height greater than the first cell height and the second cell height, wherein the third cell comprises a second plurality of active areas in a second layer below the first layer; wherein the first and second cells abut the third cell in a first direction, and the first and second cells abut each other in a second direction, wherein the first active area and a first area in the second plurality of active areas align with each other along the first direction, and first and second areas in the first plurality of active areas align with second and third areas in the second plurality of active areas along the first direction.
10. The integrated circuit of claim 9, wherein the first cell height is smaller than the second cell height and the third cell height.
11. The integrated circuit of claim 9, wherein a number of the first plurality of active areas is smaller than a number of the second plurality of active areas.
12. The integrated circuit of claim 9, wherein the second cell further comprises: a third plurality of active areas in the second layer, wherein the third plurality of active areas overlap the first plurality of active areas, and align with corresponding ones of the second plurality of active areas in a layout view.
13. The integrated circuit of claim 9, wherein the second cell further comprises a non-doped region between the first and second areas in the first plurality of active areas, and a third area in the second plurality of active areas is disposed between the second and third areas in the second plurality of active areas.
14. The integrated circuit of claim 9, wherein the first active area aligns with a fourth area in the second plurality of active areas along the first direction.
15. The integrated circuit of claim 9, further comprising: a plurality of gate stripes that extend continuously in the second direction and shared by the first and second cells.
16. A method, comprising: forming a first plurality of active areas that extend in a first row, wherein one of the first plurality of active areas overlaps another one of the first plurality of active areas in a layout view; forming a second plurality of active areas that extend in a second row, wherein one of the second plurality of active areas overlaps another one of the second plurality of active areas in the layout view; forming a third plurality of active areas that extend in a third row between the first and second rows, wherein a length of one in the third plurality of active areas and a length of one in the second plurality of active areas are different from each other; and forming a plurality of gate structures extending to cross at least one of the first plurality of active areas, at least one of the second plurality of active areas, and at least one of the third plurality of active areas.
17. The method of claim 16, wherein two overlapping active areas have different conductivity types.
18. The method of claim 16, wherein the third plurality of active areas are shorter than the first plurality of active areas and the second plurality of active areas.
19. The method of claim 16, wherein one of the third plurality of active areas overlaps another one

of the third plurality of active areas in the layout view.

20. The method of claim 16, further comprising: forming a fourth plurality of active areas that extend in a fourth row adjacent to the second row, wherein one of the fourth plurality of active areas overlaps another one of the fourth plurality of active areas in the layout view, wherein the third and fourth plurality of active areas are shorter than the first plurality of active areas and the second plurality of active areas.
